Week 03 Practice Problems (Optional)

2026-09-09

> Draft. This page is unfinished and will change.

```
using Distributions
using Printf
```

Attempt these before Wednesday. Not graded and not submitted. Open the worked answer after you have tried the problem.

The module 1 written test draws from this pool.

Likelihood of a Poisson sample

You observe the following annual counts of days below freezing in Houston: $y = (8, 5, 12, 3, 7)$. Assume the counts are independent draws from a Poisson distribution with unknown rate λ.

a. Write down the log-likelihood $\log p(y \mid \lambda)$ as a function of λ, dropping terms that do not depend on λ.
b. Find the MLE $\hat{\lambda}$ by setting the derivative to zero.
c. Compute $\hat{\lambda}$ numerically and give the probability of observing zero freezing days in a year under this estimate.

Answer

Part a. The Poisson log-likelihood for one observation is $y_i \log \lambda - \lambda - \log(y_i!)$. Summing over $n = 5$ observations and dropping the constant:

$$\log p(y \mid \lambda) = \left(\sum_{i=1}^{5} y_i\right) \log \lambda - 5\lambda + \text{const.} \tag{1}$$

Part b. Differentiate and set to zero:

$$\begin{aligned} \frac{d}{d\lambda} \log p &= \frac{\sum y_i}{\lambda} - 5 = 0 \\ \hat{\lambda} &= \frac{\sum y_i}{5} = \bar{y} \end{aligned} \tag{2}$$

The MLE is the sample mean.

Part c.

```
y = [8, 5, 12, 3, 7]
λ_hat = mean(y)
p_zero = pdf(Poisson(λ_hat), 0)
@printf("λ̂ = %.3g, P(Y=0) = %.3g", λ_hat, p_zero)
```

```
λ̂ = 7, P(Y=0) = 0.000912
```

The MLE of the rate is the sample mean. The probability of zero freezing days is vanishingly small under this estimate.

Bayes' theorem with a conjugate prior

A Gamma(α, β) prior on the Poisson rate λ is conjugate: after observing n independent Poisson draws with sum $s = \sum y_i$, the posterior is Gamma($\alpha + s$, $\beta + n$).

Using the same data as above and a prior Gamma(2, 1):

a. Write down the posterior distribution.
b. Compute the posterior mean and compare it to the MLE.
c. The prior mean is $\alpha/\beta = 2$. In which direction does the prior pull the estimate, and why?

Answer

Part a. With $\alpha = 2$, $\beta = 1$, $s = 35$, $n = 5$:

$$\lambda \mid y \sim \text{Gamma}(2 + 35, 1 + 5) = \text{Gamma}(37, 6) \tag{3}$$

Part b. The posterior mean is $(\alpha + s)/(\beta + n)$.

```
α_post = 2 + 35
β_post = 1 + 5
post_mean = α_post / β_post
@printf("Posterior mean = %.3g, MLE = %.3g", post_mean, 7.0)
```

```
Posterior mean = 6.17, MLE = 7
```

Part c. The prior mean is 2, which is below the sample mean of 7. The posterior mean is pulled slightly below the MLE toward the prior. With only 5 observations the prior has a visible effect; with 50 it would be negligible.

Method of moments versus MLE

For a Normal distribution, the method of moments and MLE both give $\hat{\mu} = \bar{y}$. For the variance, method of moments gives $s^2 = \sum(y_i - \bar{y})^2/(n-1)$ and MLE gives $\hat{\sigma}^2 = \sum(y_i - \bar{y})^2/n$.

a. For $n = 10$, by what percentage do the two variance estimates differ?
b. At what sample size does the difference fall below 1%?

Answer

Part a. The ratio is $n/(n-1)$.

```
n = 10
ratio = n / (n - 1)
@printf("MoM/MLE ratio = %.3g, difference = %.3g%%", ratio, 100 * (ratio - 1))
```

```
MoM/MLE ratio = 1.11, difference = 11.1%
```

The method-of-moments variance is about 11% larger than the MLE variance.

Part b. Solve $n/(n-1) - 1 < 0.01$, which gives $1/(n-1) < 0.01$, so $n > 101$.

```
n_threshold = ceil(Int, 1 / 0.01) + 1
@printf("n > %d", n_threshold)
```

```
n > 101
```

The difference is negligible for the record lengths used in this course.

What a bootstrap confidence interval means

You bootstrap the 100-year return level of annual minimum temperature and obtain a 95% confidence interval of [15, 45] years.

a. A colleague says "there is a 95% probability the true return period is between 15 and 45 years." Is this a correct interpretation of a frequentist confidence interval?
b. Under what framework would that statement be correct?

i Answer

Part a. A frequentist confidence interval says: if you repeated the experiment many times and built a 95% CI each time, 95% of those intervals would contain the true value. Any single interval either contains the true value or it does not. The statement "95% probability the true value is in this interval" treats the parameter as random, which the frequentist framework does not.

Part b. Under the Bayesian framework, the parameter has a distribution, and a 95% credible interval does mean "95% posterior probability that the parameter lies in this range." The two intervals often agree numerically but their interpretations differ.

Prior sensitivity

You are estimating the Poisson rate of days below freezing. Two analysts use the same data ($n = 5$, $s = 35$) but different priors:

- Analyst A: Gamma(2, 1), prior mean 2
- Analyst B: Gamma(50, 10), prior mean 5

a. Compute both posterior means.
b. Which analyst's prior has more influence, and why?

i Answer

Part a.

```
s = 35
n = 5
post_a = (2 + s) / (1 + n)
post_b = (50 + s) / (10 + n)
@printf("Analyst A: %.3g, Analyst B: %.3g, MLE: %.3g", post_a, post_b, s / n)
```

```
Analyst A: 6.17, Analyst B: 5.67, MLE: 7
```

Part b. Analyst B's prior has more influence because it carries a larger effective sample size ($\beta = 10$ prior "pseudo-observations" versus $\beta = 1$). With only 5 real observations, a prior that acts like 10 observations dominates. Both posteriors would converge to the MLE with enough data.

Bibliography